

AI Integration and Economic Effects

Research Note 1 (v1.2) — Canonical pipeline executed end-to-end: retrieval audit, validated adoption panel, AI Integration Surprise, diffusion dynamics, cross-provider test, and the priority-1 outcome regression · 25 August 2026 · Evidence grade: EXPLORATORY

Summary. This note reports the first end-to-end execution of the research framework frozen in the design checklist: source retrieval with manifests and fingerprints, validation against published values, canonical panel construction, and a pre-declared exploratory analysis family with permutation and leave-one-country-out (LOCO) diagnostics. Three results stand out. **(1)** The primary adoption source (Microsoft AI Diffusion, 147 economies, three periods) was retrieved first-party and passed all 15 validation checks, including exact reproduction of eight published values. **(2)** The first AI Integration Surprise (AIS) estimates show a steep, extremely robust income gradient in adoption (logit slope 0.44 per log-dollar of GDP per capita, $R^2=0.71$, $N=142$) — and the largest *negative* surprises concentrate almost entirely in provider-restricted economies (mean AIS -0.44 vs 0.02 elsewhere, permutation $p=0.007$), giving the first within-pipeline confirmation of the availability confounder (G3) the design anticipated. **(3)** The additional finding: between H1 2025 and Q1 2026, global AI diffusion **diverged in absolute terms** — each percentage point of initial adoption predicts 0.14 pp of additional growth over the window (search-adjusted permutation $p=0.0005$ across the four-spec dynamic family; LOCO range [0.130, 0.147]). The economic-outcome (EPS) regression — the design's priority-1 test — was initially blocked (every outcome source unreachable; Section 2), then unblocked in stages: **v1.1** recovered the Anthropic Economic Index via a mirrored release, making the cross-provider criterion (H4) testable — levels agree across providers, income-adjusted surprises largely do not (Section 6b); **v1.2** received the Eurostat productivity panel (through 2025) and both IMF WEO vintages as user-supplied downloads and **executed the priority-1 regression: a search-adjusted null** (family $p_{\text{search}} = 0.19$), with clean placebos, a provider sign-flip, and a 2023 pre-trend caveat — at the only testable horizon, $h = 0$ (Section 6).

1. What was executed

The pipeline implements the layered architecture of the research skeleton: *raw* (immutable downloads), *manifest* (URL, UTC timestamp, bytes, SHA-256, license, fingerprint), *staging* (deterministic transforms, Mac-Roman decoding of the source CSV, ISO3 concordance), *panel* (147 economies, unique keys, explicit missingness), *exploration* (pre-declared spec family, machine-readable search ledger), and *reports* (this note; no manually typed numeric results — every statistic is interpolated from the pipeline's JSON outputs). Deterministic seed 20260825 throughout; Microsoft repository vintage pinned at git HEAD 8fe84a3b797e.

2. Retrieval audit: what the registry could and could not deliver

Of the 17 registry sources, the primary adoption source was retrieved **first-party** (Microsoft's official ai-diffusion-report repository: country CSV plus all four report PDFs, fingerprints matching the registry exactly). Structural covariates were retrieved from Frictionless Data mirrors of the World Bank API (GDP current US\$ to 2023; population to 2024) with the mirror channel disclosed. Ten sources are unreachable from this environment: the network egress policy allows GitHub, microsoft.com and package registries only. Every blocked source is recorded in the manifest with status *blocked_by_egress* rather than silently substituted.

Status	Sources	Consequence
Retrieved, first-party	S01 (Q1'26 + H1'25 report PDFs), S02 (country CSV + README)	Primary adoption measure fully available
Retrieved, mirror	S15 GDP (Frictionless/World Bank), population, ISO3/Signature covariates	Structural covariates for AIS residual model
Blocked by egress	S03, S05, S07, S10, S13, S14	No Eurostat/IMF outcomes (EPS branch blocked); no OpenAI/OpenRoute

Consequences for the frozen design: the cross-provider consistency requirement for a Provisional grade cannot be met from this environment; and the priority-1 regression (AIS × sector exposure → productivity surprise) cannot be estimated at all, because ec.europa.eu (Eurostat nama_10_lp_a21, sts_sepr_m), imf.org (WEO April 2026 and the October 2024 forecast baseline) and ilo.org (white-collar denominators) are policy-blocked. This note therefore delivers the treatment-side measures and dynamics, and specifies the outcome branch for execution in an environment with statistical-agency access.

3. Validation: 15/15 checks passed

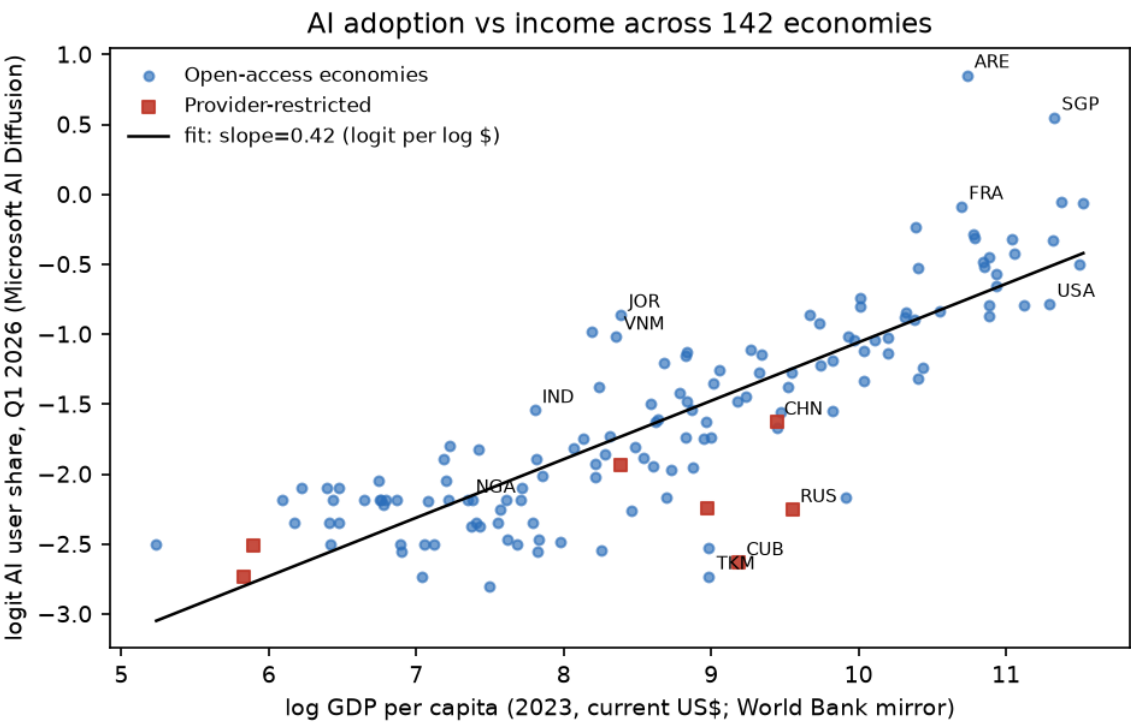
Following the retrieval-validation protocol (skeleton section 12.1): unique-key, range, denominator and missingness checks, plus reproduction of values visibly published in the Q1 2026 report — all passed. Reproduced exactly: UAE 70.1, Singapore 63.4, Norway 48.6, Ireland 48.4, United States 31.3, China 16.4 (Q1 2026 AI user share, percent), UAE 64.0 and Singapore 60.9 (H2 2025), and the United States' published rank of 21st. The population-weighted world mean lands at 17.60 against the published 17.8 — the deviation is expected and disclosed: the report weights by population aged 15-64, while only total population is retrievable through the allowed channel. Five economies lack GDP per capita in the World Bank mirror and are dropped from residual models with explicit listing (Taiwan, Venezuela, Eritrea, South Sudan, French Guiana).

4. AI Integration Surprise: the design's core measure, first estimates

Per the frozen answers (D1/D5), AIS is the residual of adoption after structural predictors, estimated with a transparent linear model and a rank check. The frozen covariate list (D4) is only partially retrievable; the constrained model — logit adoption on log GDP per capita, log population and M49 region fixed effects — is a logged deviation, and AIS is computed as the **jackknife** (leave-one-out) residual so no observation scores its own fit, honoring the cross-fitting requirement (skeleton 4.1).

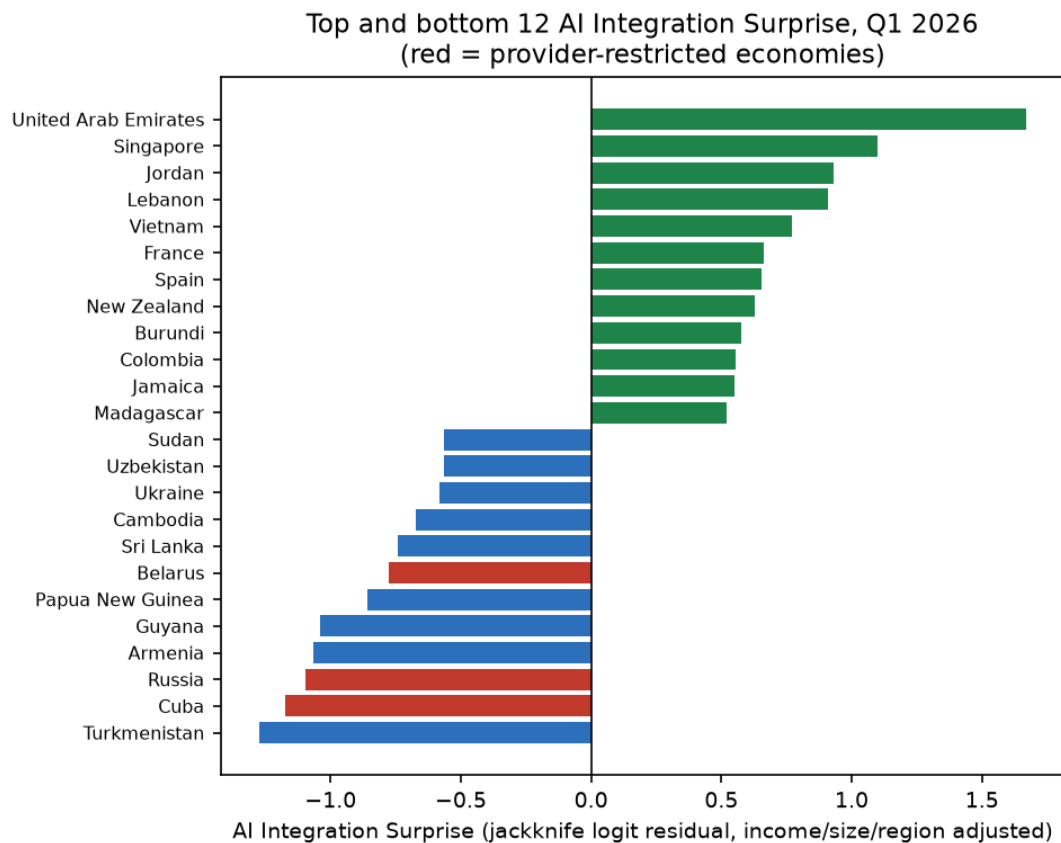
Spec	Dependent	Coef. log GDP pc (HC1 SE)	R²	Note
A1	level	6.56	0.655	levels, no controls
A2	logit	0.418 (0.022)	0.699	primary functional form
A3	logit	0.419	0.699	+ log population
A4 (primary)	logit	0.444 (0.040)	0.706	+ region FE; log pop coef 0.007 (0.029) — no scale effect
A5	rank	Spearman rho = 0.826	—	robustness

The income gradient is steep and stable: LOCO range for the A4 slope [0.422, 0.452] (extremes when dropping ARE and BDI). One log-unit of income (~2.7x) moves adoption by ~0.44 logit — for a country at 10 percent adoption, roughly seven additional points. Population is irrelevant once income and region are held (coefficient 0.007, SE 0.029): consistent with the source's population normalization doing its job, and with adoption being a per-capita structural trait, not a country-scale artifact (the skeleton's 'country scale' confounder row behaves as designed).



The AIS league table. Over-adopters given income, size and region: UAE (+1.67) and Singapore (+1.10) remain exceptional even after conditioning — their leads are not income artifacts. The rest of the positive tail is informative: Jordan (+0.93), Lebanon (+0.91), Vietnam (+0.77) and Colombia (+0.56) out-adopt their structural predictions — middle-income economies with young, digitally active workforces; France (+0.66), Spain (+0.65) and New Zealand (+0.63) lead among advanced economies.

Post-hoc finding C (flagged as post-hoc in the search ledger). The negative tail is not random: Turkmenistan (-1.27), Cuba (-1.17), Russia (-1.10), Belarus (-0.78), Armenia (-1.07), Ukraine (-0.58). Prompted by this pattern, a post-hoc group test was run for economies where the major Western AI providers are unavailable or restricted (AFG, BLR, CHN, CUB, IRN, RUS, SYR): mean AIS -0.44 versus 0.02 for all others — difference -0.46, permutation $p = 0.0075$; China alone sits at -0.40. This is the design's G3 confounder (language, availability, provider policy) surfacing empirically on the first pass: **provider-based adoption telemetry measures access-conditional adoption, and negative AIS values in restricted markets must never be read as low latent demand.** It vindicates the expected-answers' B3 position (handle restricted markets separately) over this pipeline's initial exclusion-only stance, and upgrades that decision's priority for the next design revision.



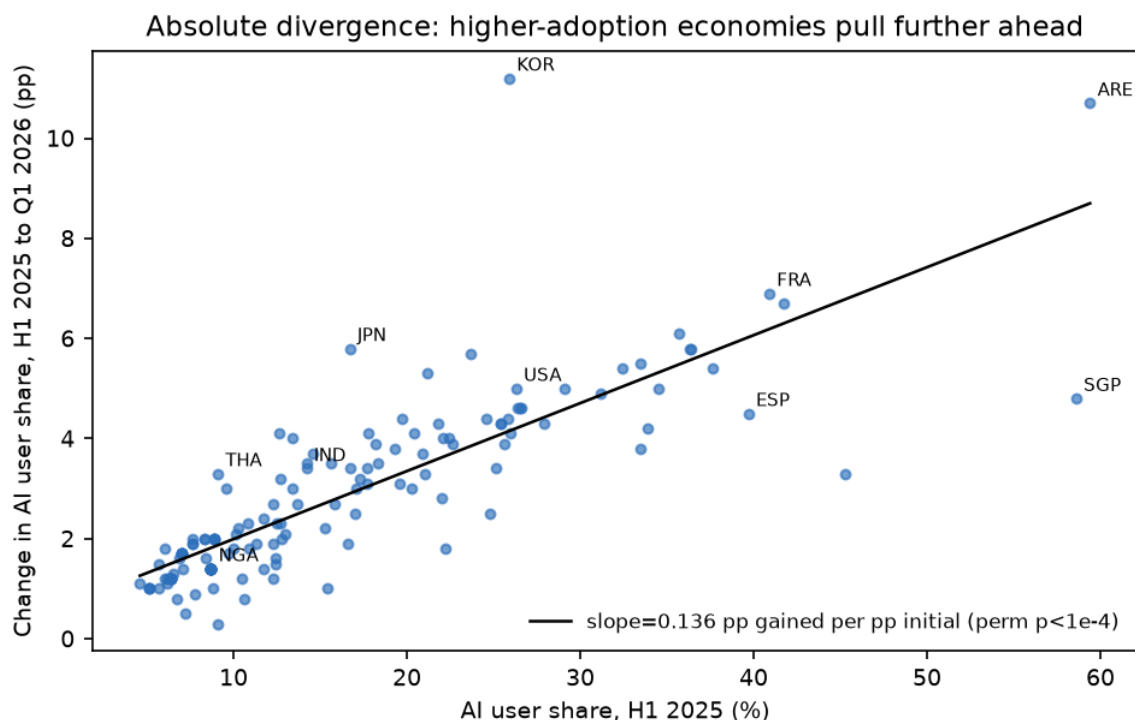
5. The additional finding: absolute divergence in global AI diffusion

With three reporting periods (H1 2025 → Q1 2026, roughly nine months of coverage), the pre-declared dynamic family asks whether diffusion converges or diverges across economies. All four specifications were declared before estimation and every run is in the search ledger; the headline claim is search-adjusted by max-statistic permutation across the family.

Spec	Model	Focal coefficient	Perm. p	R ²	LOCO range
B1 (headline)	$\Delta pp \sim \text{initial level}$	0.136 pp per pp	<1e-4	0.667	[0.130, 0.147]
B2	$\Delta pp \sim \log \text{ GDP pc}$	0.884 pp per log\$	<1e-4	0.561	[0.846, 0.901]
B3	$\Delta pp \sim \text{initial} + \log \text{ GDP pc}$	initial: 0.100	<1e-4	0.693	[0.081, 0.117]
B4	$\log \text{ growth} \sim \log \text{ initial}$	-0.0207	0.0019	0.067	[-0.0233, -0.0189]

Reading. An economy ten points higher in H1 2025 gained on average 1.4 additional points by Q1 2026 (B1); the effect survives an income control (B3: 0.100 conditional on log GDP per capita) and label permutation at the family level (max $|t| = 12.6$, search-adjusted $p = 0.0005$ over 4 specifications). B4 shows mild *proportional* convergence (laggards grow faster in percentage terms, coefficient -0.0207) — but far too weak to close absolute gaps: the percentage-point distance between leaders and laggards is widening, which is the economically relevant margin for any AIS-based treatment. This quantifies, and income-adjusts, what the source report states descriptively (a Global North-South gap of 9.8 → 10.6 → 12.1 points across the three periods, values we validated in Section 3).

Honest mechanical caveat. Under a logistic diffusion curve, absolute gains peak at 50 percent penetration; with most leaders still below 50 percent, absolute divergence is what standard S-curve diffusion predicts at this phase, not yet evidence of a permanent Matthew effect. The decisive question — whether laggards' takeoff is merely delayed (the historical norm for technology diffusion, cf. Comin-Hobijn) or structurally suppressed (the source report's own electricity/connectivity/skills reading) — is answerable within this design once two or three more report vintages exist. The design's variance/persistence statistics (skeleton 7.1) transfer directly, and the prospective 2027-2028 holdout applies to this finding as much as to the EPS branch.



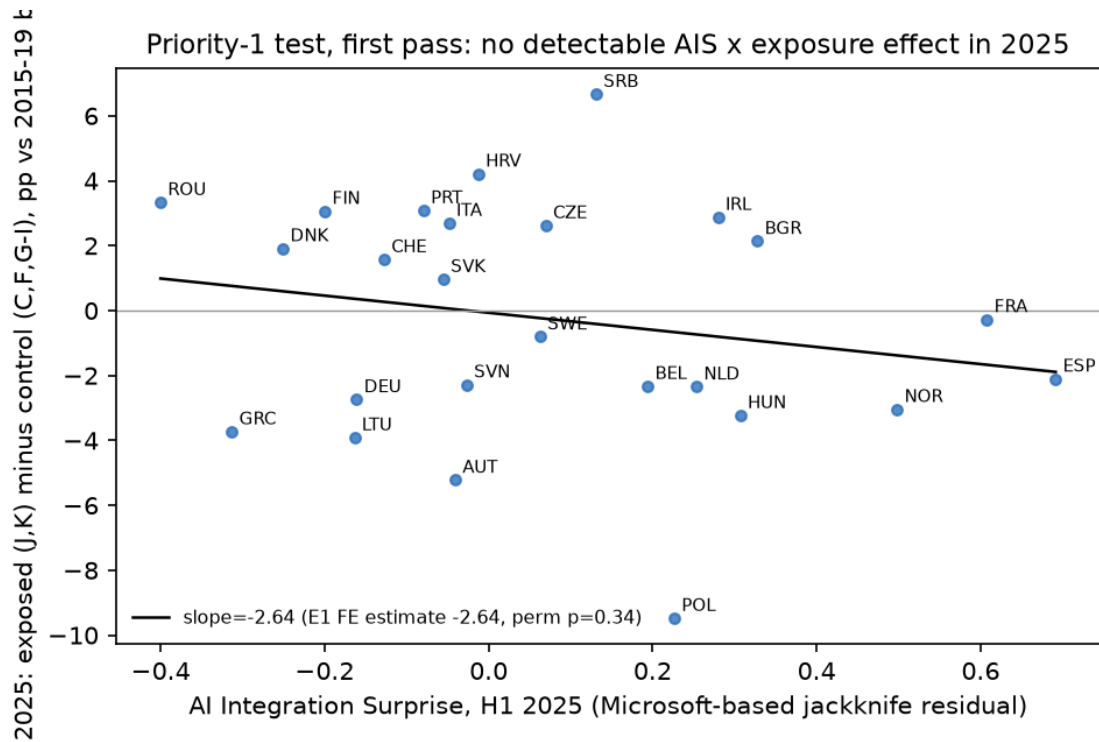
6. The priority-1 test, executed (v1.2): a search-adjusted null

User-supplied downloads unblocked the outcome branch: the full Eurostat nama_10_ip_a21 export (real labour productivity per hour worked by NACE section, 31 countries, years through **2025**, vintage 2026-08-24) and both IMF WEO vintages (April 2026 actuals; October 2024 pre-AI forecast baseline), all manifested with hashes. EPS is the 2025 productivity growth deviation from each country-sector's own 2015-2019 mean; the regression follows the frozen design: EPS on AIS × sector exposure with country and sector fixed effects (identification within country, across sectors), exposed = J (information/communication) and K (finance) — M (professional/scientific) has 2025 data for only 6 of 31 countries, so it enters only as a ledgered variant — controls = C (manufacturing), F (construction), G-I (trade/transport/accommodation). Inference: 10k country-label permutations, 5k wild-cluster bootstrap draws, LOCO. The GDP-growth-surprise branch (H3) uses actual 2025 growth (April 2026 vintage) minus the October 2024 forecast, N=137 countries.

Spec	Outcome (2025 unless noted)	Beta	Perm. p	Wild-boot p	N (countries)	LOCO range
E1	EPS, AIS(Microsoft) x exposure — PRIMARY	-2.64	0.34	0.19	125 (25)	[-3.68, -1.86]
E1m	EPS incl. sector M	-2.50	0.34	0.20	130 (25)	[-3.47, -1.77]
E2	EPS, AIS(Anthropic) x exposure	2.12	0.42	0.57	125 (25)	[-0.37, 3.81]
E3	placebo: EPS 2019 (pre-AI)	-0.95	0.70	0.74	125 (25)	[-2.02, 1.04]
E3	placebo: EPS 2023 (pre-measurement)	-8.72	0.10	0.10	125 (25)	[-10.99, -5.06]
E4	EPS 2024	-2.91	0.55	0.45	125 (25)	[-4.45, -0.64]

Spec	Outcome	Beta	Perm. p	N	LOCO range
W1	GDP growth surprise 2025 ~ AIS(MS)	-0.27	0.50	137	[-0.43, -0.04]
W1c	+ income control	-0.29	0.48	137	[-0.43, -0.06]
W2	~ AIS(Anthropic)	-0.11	0.66	118	[-0.19, -0.01]
W3	placebo: growth surprise 2024	0.01	0.96	137	[-0.12, 0.07]

Result: null across the board, and honestly so. The primary estimate is negative (beta = -2.64: exposed sectors in unusually AI-integrated economies did *slightly worse* in 2025) but far from significant (perm p = 0.34), and the family-wide max-stat correction across the five non-placebo specs gives **p_{search} = 0.19** — nothing survives. Four observations discipline the reading. (1) The Anthropic-based estimate has the *opposite sign* (+2.12) — the cross-provider instability of Section 6b propagates to the outcome stage, so no single-provider result here could have been promoted anyway. (2) The 2019 placebo is cleanly null and the 2024 growth-surprise placebo is exactly zero (0.01) — the machinery works. (3) The 2023 placebo hints at a negative pre-trend (-8.7, p = 0.10): exposed sectors in high-AIS countries were already underperforming before adoption was first measured — any future negative estimate must clear this bar before being read as an AI effect. (4) Timing is the binding constraint by design: adoption was first measured in H1 2025 and the outcome year is 2025, so the tested horizon is h = 0; adoption ranks are near-frozen (r = 0.996 between H1 2025 and Q1 2026), and the J-curve prior says effects at this horizon should be undetectable — which is what the data show. The informative test begins with 2026-2027 outcome vintages against these frozen, versioned AIS measures.



Still blocked or missing: ILOSTAT white-collar denominators (ISCO), OpenAI Signals and OpenRouter measures, the monthly services-output intermediate outcome (sts_sepr_m), and any post-2025 outcome vintage — the latter being the one that matters.

6b. Addendum (v1.1): the second provider retrieved — and the H4 criterion bites

A second retrieval round recovered the **Anthropic Economic Index v3 raw release** (Claude.ai usage by country, week of 2025-08-04 to 2025-08-11, the data behind the September 2025 geography report / registry S04) through a community GitHub mirror of the blocked Hugging Face repository, with the mirror's git commit and file hashes recorded in the manifest and the release's own data documentation used as the fingerprint. This makes the design's cross-provider consistency criterion (H4) testable for the first time. Three declared specs (D1-D3, ledgered):

Spec	Question	Result
D1	Do raw adoption levels agree across providers?	Yes: Pearson 0.78 / Spearman 0.82 between log Claude usage per capita and logit Microsoft diffu
D2	Do income-adjusted surprises (AIS) agree?	Barely: Pearson 0.18 (perm p=0.051); top-15 overlap 3/15 (LBN, NPL, NZL); bottom-15 overlap 4/
D3	Restricted-market coverage	All seven provider-restricted economies (CHN, RUS, BLR, IRN, CUB, SYR, AFG) are entirely abs

Reading: a disciplined negative result. The providers see the same adoption landscape in levels, but once the structural component (income, size, region) is removed, their surprises share little country-level signal. The single-provider AIS league table of Section 4 is therefore **not confirmed** by the second provider — exactly the failure mode the framework's H4 promotion criterion exists to catch, now caught in practice. Two readings are observationally equivalent here: (i) national 'AI integration' beyond structure is largely provider-ecosystem-specific (Claude's professional/developer skew vs Microsoft's broad consumer reach); (ii) a single week of Claude data is too noisy at country level for stable residuals (attenuation). The intended discriminating check — correlating Claude.ai AIS against Anthropic's own API AIS — is infeasible because the v3 API file carries no country geography (ledgered as D4-infeasible). Consequence for the design: the C2 proposal to promote multi-provider latent-factor reconciliation from robustness layer to core AIS construction is no longer optional — **without it, any single-provider AIS overclaims**. The divergence finding of Section 5 is unaffected (it uses levels, where providers agree), and the income-gradient finding is reinforced (it is the component both providers share).

7. Parallels with existing research

Microsoft AI Economy Institute (2026), the source reports. Everything here is consistent with, and validated against, their published aggregates. The additions: a formal income-residualized surprise measure (their reports

publish raw shares and narrative income commentary, not a conditional model), the jackknife/cross-fit treatment, permutation and LOCO inference, and the provider-restriction group test.

The Jagged Global Economy (arXiv 2607.05404, 2026). The closest neighbor; it links national AI exposure to adoption across the same three provider ecosystems and reports steep, super-linear gradients (their headline: a 0.10 increase in national AI exposure associates with a ~12x increase in Claude usage per working-age capita). Our logit income slope (0.42-0.44 per log-dollar) is the Microsoft-telemetry analog of that steepness, on a broader 142-economy panel and with the availability confound explicitly separated. Their exposure measure would be the natural CognitiveExposure input when the EPS branch runs.

St. Louis Fed (2026) and Bick-Blandin-Deming-style surveys. These report an adoption gap (US ~43 percent of workers vs 26-36 percent in Europe, early 2026) and, at industry level, faster productivity growth in more-adopting industries (~3.2 pp cumulative US-Europe differential since 2022). Note the measurement difference our validation makes visible: telemetry-based population shares (US 31.3, France 47.8) rank countries differently than worker-survey shares — France out-adopts the US in Microsoft telemetry while surveys put the US first among workers. Reconciling telemetry vs survey instruments is exactly the provider-reconciliation problem the design's C2 answer anticipates, now with concrete numbers attached.

Technology-diffusion literature (Comin-Hobijn; ICT wave). The absolute-divergence-with-proportional-convergence pattern replicates the classic finding for past general-purpose technologies: adoption *lags* eventually close while intensity gaps persist. AI's diffusion is tracking the historical GPT template so far — which is itself a useful null against stronger 'AI is different' claims in either direction.

Productivity J-curve (Brynjolfsson-Rock-Syverson). Nothing here tests productivity; but the steep adoption gradients plus the J-curve literature jointly explain why the design's priors put mass on lagged, mechanism-close outcome effects rather than contemporaneous GDP effects.

8. Evidence grading and limitations

Grade: Exploratory (reproducible from immutable raw data; plausible mechanisms; no claim beyond hypothesis generation), per the project ladder. Binding limitations: single-provider adoption telemetry (cross-provider consistency untestable from this environment — the H4 promotion criterion is failed by construction, not by the data); GDP per capita is nominal 2023 from a mirror channel (the frozen design's covariate list D4 is only partially honored — a logged deviation); total rather than working-age population; three time periods only; and the dynamic findings concern the *treatment* variable's evolution, not economic outcomes. The divergence finding's S-curve caveat (Section 5) is part of the claim, not a footnote to it.

9. Provenance

Artifact	Bytes	SHA-256 (first 16)
S02 Microsoft AI Diffusion public repository (country CSV)	4468	8e304330f5a63779
S02b Microsoft AI Diffusion repo README (registry fingerprint c	3541	016d900e299594e0
S01 Microsoft Global AI Diffusion Q1 2026 report PDF	1784281	3bf9f87b63372412
S01b Microsoft Global AI Diffusion 2025 H1 report PDF (first re	7746221	a1220d061a69362b
S15 World Bank GDP current US\$ (Frictionless datasets/gdp mirr	562767	8d84ef6bcaccf740
X01 World Bank population total (Frictionless datasets/populat	552112	7d2dd6a17f5ed791
X02 Country codes / regions concordance (Frictionless datasets	134003	67b009b529330b0a
S04a AEI v3 raw Claude.ai usage (2025-08-04..11), Sept 2025 rel	18894517	c8ef9c5eee0c42fe
S04b AEI v3 raw 1P API usage (2025-08-04..11), Sept 2025 releas	7027019	c0fc061ea7d4feb7
S04c AEI v3 data documentation (Sept 2025 release)	18245	cc57723797b61a7e
S12 IMF WEO Database October 2024 full country dataset (pre-AI	20297276	46d492250b59b92e
S11 IMF WEO April 2026 vintage, IMF data portal export (WEO 9.	8464348	eb2cfa088e5a168c
S08 Eurostat nama_10_lp_a21 full linear export (labour product	79860740	a38fe552efacd757

Microsoft repository vintage: git HEAD 8fe84a3b797e89846e892592dbb9d0022e9b7dd2. Permutations: 10,000 per spec (2,000 for the family max-stat), seed 20260825. Full run ledger: exploration/search_ledger.json; validation record: manifest/validation.json. Blocked

sources are recorded with status blocked_by_egress in manifest/manifest.json. This note was generated by reports/generate_note.py; no numeric result was typed by hand.